

Scale transition theory.

So far:

Scale transition from individuals $\rightarrow$ population

- 1.) individuals identical + linear dynamics
+ demographic stochasticity
- 2.) individuals heterogeneous + linear dynamics.
+ demographic stochasticity

$$\frac{dE[N]}{dt} = (b-m)E[N]$$

linear in N

$\propto N$
linear.

Outcome: mean dynamics at population scale is same as a model with no variation or stochasticity all.

Corresponding model with no stochasticity:

$$\frac{dN}{dt} = (b-m)N$$

quick derivation of no stochasticity

$$\frac{dN}{dt} \frac{1}{N} = b - m$$

For a difference in expected dynamics between the variation and no-variation case we need two ingredients.

- 1.) Some process that causes variation.
(we already have this: demographic stochasticity + demographic heterogeneity + ... environmental stochasticity)

- 2.) Nonlinear dynamics.

Key ingredients to scale transition: nonlinearity + variation.

Example.

Density - dependent death rate.

$$\begin{aligned} m_i &= m + \sum_j^N c \\ &= m + cN \end{aligned}$$

Deriving a bookkeeping update rule for the IBM to avoid summing over the population

Conceptually, when a birth or death event occurs, this now changes the population state in a way that changes the death probability of every individual.

$$\begin{aligned} m_intensity(N) &= \sum_i^N m_i \\ &= \sum_i^N (m + cN) \\ &= N(m + cN) \\ &= mN + cN^2. \end{aligned}$$

birth event

$$\begin{aligned} m_intensity(N+1) &= \sum_i^{N+1} m_i \\ &= \sum_{i=1}^{N+1} (m + c(N+1)) \\ &= m(N+1) + c(N+1)^2 \\ &\dots \\ &= \underbrace{mN + cN^2}_{m_intensity(N)} + \underbrace{m + c(2N+1)}_{\text{update}} \end{aligned}$$

Deterministic model

$$\frac{dN}{dt} = b - (m + cN)$$

$$\begin{aligned}\frac{dN}{dt} &= (b-m)N - cN^2 \\ &= rN - r \frac{N^2}{K} \\ &= rN \left(1 - \frac{N}{K}\right)\end{aligned}$$

$$K \equiv \frac{r}{c} = \frac{b-m}{c}$$

Solution: $N(t) = \frac{K N_0}{N_0 + (K - N_0) e^{-rt}}$

Stochastic model: expectation

$$\frac{dE[N]}{dt} = rE[N] - cE[N^2]$$

key: $E[N^2] \neq E[N]^2$

$$E[N^2] = E[N]^2 + \text{Var}(N)$$

Scale transition step.

$$\frac{dE[N]}{dt} = rE[N] - cE[N]^2 - c\text{Var}(N)$$

deterministic
model
(no variation)

scale
transition
term
corrects for variation.

More intuition
At "equilibrium"

$$0 = rE[N] - cE[N]^2 - c\text{Var}(N).$$

$$E[N] \hat{=} K - \frac{1}{K} \text{Var}(N).$$